P7 TERM ONE LESSON NOTES

TOPIC: **PHYSICAL FEATURES IN AFRICA**

SUBTOPICS:

SESSION	COMP	CONTENT	METHODS	RELEVANCE	LESSON ACTIVITY	SCENARIOS
1.	Identifies major physical features on the map of Africa Locates the major physical features on the map of Africa	Physical features are features of the earth's surface that give it shape. These features are categorized into; drainage and relief features. **Relief features** These are landforms which are identified according to their altitude. Altitude is the height above the sea level. Altitude is measured using an instrument called an altimeter. Relief is the physical appearance of the land. Relief features are the highlands and lowlands of an area. Relief gives shape to the world. Relief features include; Mountains, Plateaus, Rift valleys, Valleys, Coastal plains, Basins, Highlands, Low lands **Drainage features**	Guided discovery Observation Brainstorming Question and answer	To help the learner understand the natural environment and how humans interact with it To help the learner value natural resources, protect wetlands,	Identifying major physical features on the map of Africa Locating major physical features on the map of Africa Locating the physical features of Africa	A child walking to school Tourist visiting east Africa Building a house Village boundary Children playing on inclined plains Family growing crops

	Locates the Physical features of Africa region by region Draws the map of Africa showing	Drainage features are landforms that contain water. Drainage features are the water bodies of an area. Drainage features include; Lakes, Oceans, Seas **Note:-** Africa's plateau is tilted to the North. This results into the northward flow of river Nile from lake Victoria towards the Mediterranean sea. The flow of river Nile from lake Victoria to lake Kyoga shows that lake Victoria is at a higher altitude than lake Kyoga. **Importance of physical features** They earn income through attracting tourists. They are used when locating places. Some of them are mining grounds. Some of them are fishing grounds. They modify climate by helping in rain formation. Some physical features form natural boundaries between countries. This is because, these features are permanently fixed.eg, River Ruvuma creates a natural boundary between Tanzania and Mozambique.	Guided group discussion	forests and water bodies To help learners understand how countries are divided using landmarks such as mountains, lakes and rivers To help learners understand their environm	region by region Drawing the map of Africa, showing major physical features Drawing diagrams illustrating the formation of major physical features in the regions of Africa	Schools location Crossing a swamp Searching for network Driver choosing a route Field walk A farmer storing food A family collecting clay

	major physical features Discusses the formation of major Physical features in the others regions of Africa	River Kagera forms a natural boundary between Uganda and Tanzania. River Semliki, mountain Rwenzori, the Western Rift valley, lake Albert and lake Edward create a natural boundary between Uganda and DRC. Mountain Elgon creates a natural boundary between Uganda and Kenya. Map of Africa showing physical features		ent ,community and country better		
2.	Defines a mountain as a large raised piece of land higher than a hill. Identifies the three types of mountains in Africa (volcanic, fold, block). Explains volcanicity in simple terms as	**MOUNTAINS IN AFRICA** A mountain is a large raised piece of land, usually higher than a hill. **Types of mountains in Africa.** Volcanic mountains • Fold mountains Block mountains **VOLCANIC MOUNTAINS.** These are mountains formed as a result of <u>volcanicity.</u> <u>Volcanicity</u> is the process by which magma is forced onto the earth's surface through a vent.	Guided discovery Observation Brainstorming	To help the learners appreciate the role of mountains in water catchment and conservation of forests	Identifying major Mountains on the map of Africa	Climbing a hill to get network A long shadow (shows mountain height and visibility) A child walking to school

	the process where hot molten rock comes out of the earth. Differentiates between magma and lava (magma is inside the earth; lava is on the surface). Describes how volcanic mountains are formed through volcanic eruptions.	Volcanicity can also be called volcanic activity or volcanic eruption. A volcano is a mountain with a vent through which magma is forced through the earth crust and onto the earth's surface. Magma refers to the molten rock in the earth crust. Lava refers to the molten rock on the earth's surface. **Types of volcanoes** Table showing types of volcanoes, examples and descriptions Other volcanic mountains in Africa include; Drakensberg mountains in Lesotho and South Africa Mountain Longonot in Kenya. Mountain Mufumbiro in Uganda. **A diagram showing the formation of a volcanic mountain** **Importance of volcanicity** It leads to formation of fertile volcanic soils which are favourable for crop growing. Volcanic mountains attract tourists who bring in income. Dangers of volcanicity.	Question and answer Guided group discussion	To help the learner recognize mountains as natural boundaries between countries To help learners understand how altitude affects climate, vegetation and human activities To help leaners know the importance of mountains	Locating major Mountains on the map of Africa Locating the Mountains of Africa	A school trip Farmers terracing a mountain A community protecting forest on the slopes

	Identifies examples of volcanic mountains in Africa such as Mt. Mufumbiro, Mt. Longonot, and Drakensberg. States the importance of volcanicity, such as fertile soils and tourism. Identify dangers of volcanicity, such as destruction of property and loss of life.	It leads to death of people and animals. It leads to destruction of property. It leads to air pollution. **Other features formed as a result of volcanicity.** Crater lakes, Calderas, Inselbergs An inselberg is an isolated hill that stands above the general level of the surrounding land e.g. Labwor hill in Karamoja, Musaijamukuru hill in Hoima, Osukuru hill in Tororo. These inselbergs attract tourists who bring in income and are also a major source of minerals eg. The Osukuru hills which provide limestone used for making cement. Lava dammed lakes Hot springs /geysers eg. Kitagata hotsprings in Bushenyi, Sempaya hotsprings in Bundibugyo Hot springs attract tourists and also help in generation of geothermal energy. **FOLD MOUNTAINS** Fold mountain are formed as a result of folding.			region by region	

	Mentions other features formed by volcanicity, such as crater lakes, calderas, inselbergs, hot springs, and lava-dammed lakes. Gives examples of inselbergs and hot springs in Uganda. Explains how fold mountains are formed when land masses are pushed together.	Folding is the process by which land mass is forced to collide due to compressional forces. Fold mountains are formed when two land masses are forced to collide by horizontal earth movements causing the rocks to be folded and uplifted. The raised parts of a fold mountain are called anticlines. The sunken parts of a fold mountain are called synclines. A diagram showing the formation of a fold mountain **Examples of fold mountains in Africa.** Cape ranges in South Africa. Atlas mountains in North West Africa (Morocco) **BLOCK MOUNTAINS** These mountains are also called horst mountains. Block mountains were formed as a result of faulting. Faulting is the cracking/ breaking of rocks in the earth's crust.			Drawing the map of Africa, showing major Mountains	

	Identify examples of fold mountains like Atlas Mountains and Cape Ranges. Explaining how block mountains are formed when the earth's crust cracks and the middle block rises. Identifying examples of block mountains such as Mt. Rwenzori, Usambara, and Pare mountains.	The central block between the two faults is lifted up by compressional forces forming a block mountain. **Forces that led to the formation of a block mountain** Compressional forces, Tensional forces. **A diagram showing the formation of a block mountain due to Compressional forces.** **Examples of block mountains in Africa.** Mountain Rwenzori in Uganda Mountain Pare in Tanzania Mountain Usambara in Tanzania Mountain Danakil in Ethiopia Mountain Uluguru in Tanzania **Highest peaks of major mountains in Africa and their location and description**			Drawing diagrams illustrating the formation of major Mountains in the regions of Africa	

3.	Defines a highland as an area made up of hills and mountains. Identify examples of highlands in Africa such as Kenyan Highlands, Guinea Highlands, Adamawa Highlands, and Ethiopian Highlands. Explains the importance of highlands, such as rain formation, fertile soils, tourism, and cool temperatures. Mentions mountains that act as natural	**HIGHLANDS IN AFRICA** A highland is an area of land consisting of hills and mountains. **Examples of highlands in Africa.** Kenya highlands, Guinea highlands, Ethiopian highlands, Adamawa highlands **Importance of highlands (hills and mountains)** They help in rain formation. They have fertile soils which support crop cultivation. They attract tourists who bring in income. They have cool temperatures which favour dairy farming. They are used by telecommunication companies to place on masts. Some mountains form natural boundaries between countries. e.g. Mountain Rwenzori creates a natural boundary between Uganda and DRC. Mountain Elgon creates a natural boundary between Uganda and Kenya. Mountain Mufumbiro creates a natural boundary between Uganda and Rwanda.	Guided discovery Observation Brainstorming Question and answer Guided group discussion Picture interpretation	To enable leaners understand why highlands have cooler temperatures that lowlands To enable learners know why many town are built in highlands To enable learners appreciate highlands as important tourist attractions	Identifying major Highlands in Africa Drawing diagrams illustrating the formation of Orographic rainfall	Farmers terracing a mountain Fresh vegetables from highland gardens (shows highlands supports specific crops) A taxi struggling uphill Highland tourism School playground on a slope

	boundaries, e.g. Mt. Rwenzori (Uganda–DRC), Mt. Elgon (Uganda–Kenya). Describes challenges caused by highlands, including poor transport, soil erosion, and landslides. Explains why transport is difficult in highland areas due to steep and rugged slopes. States why agricultural mechanisation is limited in highlands (tractors can't operate on steep slopes).	**A diagram showing the formation of Relief (Orographic rainfall)** **Disadvantages of mountains** They hinder agricultural mechanisation. They make road and railway construction difficult. Some volcanic mountains erupt and destroy property. They are greatly affected by landslides and soil erosion. **Problems faced by people living on slopes of mountains.** Severe soil erosion, Poor ground transport, Landslides. Poor agricultural mechanization **Why transport is very poor in highland areas.** It is difficult and expensive to construct roads in mountainous areas. **Note:** People living in highland areas (Kapchorwa) rear donkeys to use them as means of transport. Road transport can also be improved in highland areas by constructing winding roads. Landslides can be reduced in highland areas through planting trees		To enable learner explain high highlands support human activities To help the learners recognize highlands as sources of many rivers To help leaners understand common landscapes in Africa since the continent		Banana and coffee gardens

	Identifies ways of improving transport in highland areas (constructing winding roads, using donkeys). Explains how landslides can be reduced by planting trees. Defines a plateau as a raised flat-topped land. States that most of Uganda lies on a plateau (200–2000 metres above sea level). Identifies features found on a plateau, such as lakes, streams,	**Why agricultural mechanisation is poor in mountainous areas.** The steep slopes make the use of tractors difficult. **PLATEAUS/PLATEAUX** A Plateau is a raised flat topped piece of land. The plateau covers the largest part of Uganda. It lies between 200m and 2000m above the sea level. **Features found on a plateau** Lakes , Streams, Plains, Rivers, Hills **Economic activities carried out in plateau areas** Crop cultivation, Mining, Industrialisation, Tourism, Animal rearing, Lumbering, Fishing **A diagram showing the formation of Cyclonic / Frontal rainfall** **Major plateaus in Africa** Nyika plateau in Kenya, Bie plateau in Angola, Ahaggar plateau in Algeria, Yatta plateau in Kenya, Fouta Djallon in Guinea, Jos plateau in Nigeria, Tibesti in chad		is largely a plateau To help learners recognize plateaus as areas with moderate climate that support human life To help learners explain why plateaus are good for pastoralism		

	plains, hills, and rivers. Mentions economic activities done on plateaus (crop cultivation, mining, tourism, fishing, lumbering, industries, animal rearing). Identifies major plateaus in Africa such as Nyika, Jos, Yatta, Ahaggar, Tibesti, and Fouta Djallon.					
4.	Defines a rift valley as a long, wide depression with steep sides. Describes escarpments as the	**THE AFRICAN GREAT RIFT VALLEY** A rift valley is a long and wide depression on the earth's surface with steep sides. The steep sides of a rift valley are called escarpments. The rift valley was formed as a result of faulting.	Guided discovery Observation	To help learners understand how the earth's crust moves, forming	Identifying the countries drained by the Great rift valley	Valleys in the community Houses built far from the escarpment

	steep sides of a rift valley. Explains that the Great Rift Valley was formed by faulting (breaking of the earth's crust). Identifies the countries crossed by the African Great Rift Valley, from Syria to Mozambique. State that the rift valley is 30–100 km wide. Mentions economic activities carried out in rift valley areas (crop growing, fishing, tourism, mining, animal rearing).	The Great rift valley starts from Syria and runs through Jordan, Red Sea, Ethiopia, Kenya, Uganda, Tanzania and ends at Beira in Mozambique. The width of the Great rift valley varies from 30 to 100km. **A diagram showing the formation of a rift valley** **Arms/branches of the African Great Rift Valley, major escarpment and examples of lakes** **Economic activities done in Rift valley areas** Crop cultivation, Tourism, Animal rearing, Mining, Fishing **Dangers / disadvantages of the Rift valley.** It is affected by soil erosion. It hinders road and railway construction. It experiences very high temperatures due to low altitude. **BASINS**	Brainstorming Question and answer Guided group discussion	deep depressions To help ,earners appreciate rift valleys as important for tourism, agriculture and mineral exploration	Drawing the diagram showing the formation of a rift valley Identifying the branches of the African Great rift Valley	Banana and coffee gardens A river changing direction Playing on a flat sand beach Fishing boats parked on the shore Coconut and palm trees growing easily Tourists relaxing on a beach Ships docking at the port

	Identifies dangers of rift valleys, such as soil erosion, transport challenges, and high temperatures. Defines a basin as a large low-lying land between highlands and plateaus. Explains that basins are formed by down-warping or sinking of land. States that lakes can form in basins when water collects in them. Names examples of major African basins Defines a coastal plain as a narrow	A Basin is a large low lying land between plateaus and highlands. Basins are formed as a result of sinking of land masses/ down warping. When water collects in some basins, lakes are formed. **Examples of basins in Africa** The Victoria basin, The Congo basin, The Okavango basin, Orange basin, Senegal basin The Chad basin, The Nile basin, The Zambezi basin **COASTAL PLAINS** A Coastal plain is a narrow strip of land along the coast. Coastal plains favour the construction of sea ports and also promote tourism. The coastline of Africa is not regular due to the effect of waves. **Features found on the coastal plain** Coral reefs, Beaches, Cliffs, Lagoons Coral reefs are hard rocks at the bottom of the sea formed by polyps. **Importance of coral reefs**				

	strip of land along the coast. Explains that coastal plains support seaport construction and tourism. States that Africa's coastline is irregular due to wave action. Identify features found on a coastal plain, such as coral reefs, beaches, cliffs, and lagoons. Explains coral reefs as hard sea rocks formed by polyps. Mentions the importance of coral reefs, including tourism, jewellery making, and providing	The attract tourists who bring in income. They also provide limestone, which is used for making cement. They are used for making jewellery.		To enable learners recognizes coastal plains as areas where many major towns and markets develop To enable learners understand the importance of coastal plains in international trade and transport		

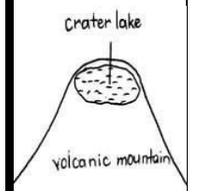
	limestone for cement.					
5.	Defines a lake as a large depression on the earth's surface where water collects. Identifies major lakes in Africa, e.g., Lake Victoria, Tanganyika, Malawi, Chad, etc.	**LAKES IN AFRICA** A lake is a large depression / hollow on the earth's surface in which water collects. **Major lakes in Africa** Lake Victoria, Lake Tanganyika, Lake Turkana, Lake Chad, Lake Tana, Lake Malawi, Lake Ngami, Lake Volta, Lake Nasser, Lake Kainji, Lake Kariba **Activity:** Locate the major lakes and rivers on the map of Africa in your book **TYPES OF LAKE FORMATIONS** Basin lakes/ depression lakes/ down warped lakes, Rift valley lakes, Lava dammed lakes, Ox-bow lakes, Manmade lakes, Crater lakes, Lagoon lakes, Glacial lakes Table with diagrams of lake formations, their descriptions and examples	Guided discovery Observation Brainstorming Question and answer Guided group discussion Picture interpretation	To help learners understand the different types of lake formation To enable learners appreciate lakes as important sources of fish, water ,transport and tourism To enable the learners learn about lake	Identifying the major lakes in Africa	Fetching water Fishermen returning with fresh fish Children playing near the shore Flooded gardens during heavy rains Buying fresh fish from the market Animals coming to drink water Farmers growing

	Explains the different types of lake formations, such as basin lakes, crater lakes, man-made lakes, and ox-bow lakes. States characteristics of basin lakes, including being wide, shallow, fresh-water, and having inlet and outlet rivers.				related challenges such as overfishing, pollution and floods To help learners explain how lakes affect climate, making nearby areas wetter and cooler	Drawing the map of Africa showing the major lakes in Africa	crops near lakes Rainwater collecting in low-lying area Using lakes for water and livelihoods Fishing Salt mining Village market near a lake Water shortage

(i) BASIN LAKES

Diagram	Description	Examples
	These are also called depression lakes/ down warped lakes. Basin lakes were formed as a result of <u>down warping.</u> <u>Characteristics of basin lakes</u> They are irregular in shape. They are wide and usually shallow. They have inlet and outlet rivers. They have fresh water. Note:	• Lake Victoria • Lake Kyoga • Lake Chad • Lake Amboseli • Lake Ngami in Botswana

	Describes crater lakes as lakes formed on volcanic mountains due to volcanicity. Explains man-made lakes as lakes created when dams are built across rivers.			• *Basin lakes have fresh water because they have outlet rivers.* • *Lake Victoria referred to as an interterritorial lake because it is shared by three East African countries.* • *Lake Victoria has got inland ports which handle imports and exports of East African countries. This is how it promotes inter-territorial trade in the region.*			Drawing diagrams of different types of lake formations	when the well dries Animal movement in search for water Digging dams in a swamp People fetching water for home use Tourists hiking around mountain slopes
			(ii) CRATER LAKES					

	Describes ox-bow lakes as lakes formed when meanders are cut off from a river. Gives examples of each type of lake formation in Africa (e.g., L. Katwe – crater; L. Volta – man-made). Explains why Lake Victoria is an interterritorial lake (shared by three			These are formed on top of volcanic mountains. They are formed as a result of volcanicity.	• L. Katwe in Uganda. • L. Nyamunuka in Uganda • L. Muhavura in Uganda. • L. Panjam in Nigeria				
			(iii) MAN-MADE LAKES	They are formed as a result of <u>dam construction.</u> They can also be formed as a result of digging valley dams in swamps.	• L. Volta in Ghana (the largest) • L. Nasser on R.Nile in Egypt. • L. Kariba in Zambia				

	East African countries). Recognises that some lakes promote trade, fishing, HEP and tourism.	• L. Kainji on R.Niger in Nigeria **VII) OX-BOW LAKES** These are formed as a result of river deposition. They are mainly formed on meandering rivers in the lower stage of a river. A diagram showing the formation of ox-bow lakes. **Examples of ox-bow lakes** Lake Utange on river Rufiji Lake Manzala on the Nile delta in Egypt Lake Gambi on river Tana in Kenya.				
6.	Defines a river as a mass of flowing water on the earth's surface.	**RIVERS IN AFRICA** A river is a mass of flowing water on the earth's surface.	Guided discovery Observation	To enable learners understand how rivers are formed and how	Identifying the major rivers in Africa	Flooded gardens during heavy rains

	Identifies major rivers in Africa, such as River Nile, Congo, Zambezi, Volta, Senegal, Gambia, Limpopo, Rufiji, etc. States that most rivers in Africa originate from highlands and plateaus because they receive a lot of rainfall.	Most rivers in Africa originate from high plateaus and highlands because such areas receive plenty of rainfall which provides water to the rivers. Rivers flow from areas of high altitude to areas of low altitude. Rivers continuously get their water from rainfall, melting snow or underground water flow. **Examples of major rivers in Africa.** River Nile, River Zambezi, River Gambia, River Congo, River Volta, River Benue, River Ruvuma, River Senegal, River Atbara, River Kagera River Orange River Tana, River Rufiji, River Limpopo **Examples of seasonal rivers in Africa.** River Turkwel in Kenya River Lagh Bogal in Kenya River Lagh Bor in Kenya Terms related to rivers. A river source is a point where a river begins to flow. A river mouth is a point where a river ends its flow. A tributary is a small river that joins the main river. A distributary is a small river that branches away from the main river. A river confluence is a place where two or more rivers meet.	Brainstorming Question and answer Guided group discussion Picture interpretation	they flow from highlands to lowlands To help learner describe the importance of rivers in irrigation, hydropower, transport and fishing To help learners recognize rivers as natural boundaries between countries		A child fetching water from a stream Farmers irrigating their crops Fishing as a source of income A bridge used to cross a river Tourism at waterfalls Transporting goods in a boat Generation of hydroelectricity

	Explains that rivers flow from high altitude to low altitude. Describes sources of river water (rainfall, melting snow, underground water). Explain meanings of key river terms Describes features formed by rivers, such as waterfalls,	An estuary is a wide part of a river as it ends its flow/enters the sea or a lake. A drainage basin is an area of land drained by a river, its tributaries and distributaries. A flood plain is a flat area near a river that often floods when the water level rises. A delta is the place where a river divides/splits into several streams as it ends its flow. E.g.-R.Nile, R.Zambezi, R.Okavango and R.Niger form delta at their mouths. Note: Deltas are formed at the mouths of rivers, as a result of river deposition. Deltas have rich fertile soil for crop growing. They also attract tourists who bring in income. A waterfall is a steep flow of a river. Waterfalls attract tourists who bring in income, and also help in generation of hydroelectricity. However, waterfalls may cause water accidents and also block water transport. **A diagram showing a river system.** **Stages of river development**		To enable the learner learn how rivers contribute to soil fertility through flooding To help learners appreciate rivers as water sources for domestic use and wildlife		Pottery

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<td>deltas, estuaries, gorges, flood plains and ox-bow lakes.

Identifies that deltas are formed by river deposition at the river mouth.</td>
<td>A river has three main stages/courses
These are;
The Youth stage/Upper course
The Old stage/ Lower stage (Senile)
The Mature stage/ Middle stage
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<tr><th>Stage</th><th>Characteristics</th></tr>
<tr><td>The Upper course</td><td>It has waterfalls and rapids
The river flows fast.
It forms a V-shaped valley (gorge).
There is a lot of vertical erosion.
NOTE:
This stage is suitable for generation of hydro electricity</td></tr>
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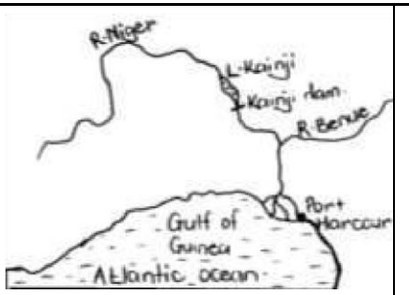
	Describes the three stages of a river States the importance of waterfalls (tourism, HEP) and their dangers (water accidents, block transport).			because it has many waterfalls. The upper stage is suitable for tourism because it has waterfalls which attract tourists.					
			The Middle stage	The river flows gently. It forms a U-shaped valley. There is lateral erosion in this stage. It begins to form meanders.					
			The Lower stage	The river flows slowly.					

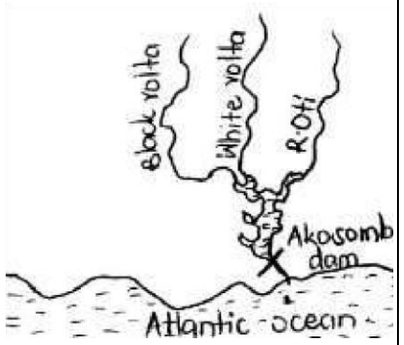
			It forms meanders and ox-bow lakes. It form a deltas or an estuary. The river deposits its load forming flood plains.				
		Features found along river valleys. Waterfalls, Deltas, Estuaries, Gorges, Ox-bow lakes					
7.	States that River Nile is the longest river in the world. Identifies the local name of River Nile as "Kiira." Explains why River Nile flows northwards (because the north is at a lower altitude).	✦ River Nile is the longest river in the world. ✦ Its local name is Kiira. ✦ It flows northwards because the North is on a low altitude.	Source Mouth Countries drained Main tributaries	-Lake Victoria -Mediterranean sea -Uganda -Sudan -Ethiopia -Egypt -South Sudan -Blue Nile from lake Tana in Ethiopia -River Atbara -River Achwa in Uganda	Guided discovery Observation Brainstorming	To help leaners understand the longest river in Africa and the world To enable the learner appreciate the role of river	Identifying the countries drained by river Nile	Generation of hydroelectricity Fetching water for domestic use Irrigation farming

	Identifies Lake Victoria as the source and the Mediterranean Sea as the mouth of River Nile. Names the countries drained by River Nile: List the main tributaries of River Nile Identifies major waterfalls on River Nile Names major hydropower dams along the Nile	✦ John Hanning Speke was the first European explorer to see the source of river Nile.			Question and answer Guided group discussion Picture interpretation	Nile in settlement patterns along the Nile valley To enable the learner recognize the role river Nile in irrigation especially in Egypt and Sudan To help learners learn its Importance in hydroelectric power To help learners learn		Using a bridge to cross Fishermen earning a living Using water to cool industrial machines Flooding in the Nile valley Tourists visiting waterfall and the source of river Nile Recreation activities
			Waterfalls	-Karuma falls -Murchison falls -Itanda falls				
			Hydroelectric power dams	-Nalubaale power dam -Kiira dam in Uganda -Bujagali power dam -Karuma dam -Aswan high dam in Egypt -Jebel Aulia dam in Sudan -Roseires dam in Sudan Note:				

	Explains that Aswan High Dam led to the creation of Lake Nasser. Explains why the Nile is called the "lifeblood of Egypt." Identifies Nile Valley countries and explain that the Nile Valley is the land drained by the Nile and its tributaries. Explains why the Nile Valley is densely populated (fertile soils &				The construction of Aswan High dam resulted into creation of lake Nasser. The Nile is the life blood of Egypt because the people of Egypt depend on the water of the Nile for irrigation and domestic use.			about internatio nal relations since river Nile flow through many countries		
				Nile valley countries	-Uganda,-South Sudan, -Ethiopia, -Sudan, -Egypt. <u>NB:-</u> *<u>The Nile valley</u> is the area drained by river Nile, its tributaries and distributaries.* *The Nile valley is densely populated because it receives reliable rainfall, and also has fertile soils which support crop growing.*					

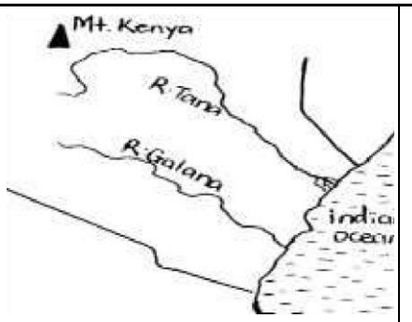
	reliable water supply).						
8.	States that River Congo is the biggest river in Africa (by volume of water). Identifies that River Congo forms an estuary at its mouth. Explains that River Congo forms a natural boundary between DRC and Congo-Brazzaville. Names major power dams on River Congo	RIVER CONGO	✞ It is the biggest river in Africa. ✞ It forms an estuary at its mouth. ✞ It forms a natural boundary between DRC and Congo-Brazzaville <u>Power dams on river Congo</u> -Nzilo power dam -Inga power dam <u>Major tributaries of river Congo</u> -River - River Kasai ✞ It carries the largest volume of water to the ocean. because; -it has many tributaries -it originates from equatorial climatic region which receives plenty of rainfall throughout the year.	Guided discovery Observation Brainstorming Question and answer Guided group discussion Picture interpretation	To enable the learner understand one of the deepest and voluminous rivers in Africa To help learners appreciate how river Congo influence the climate of central Africa To help the learner	Identifying the main tributaries of river Congo	Generation of hydroelectricity Fetching water for domestic use Irrigation farming Using a bridge to cross Fishermen earning a living Using water to cool industrial machines

	Mentions major tributaries of River Congo States that River Congo carries the largest volume of water to the ocean and explain why:			learn how river Congo supports rainforests and wildlife		Flooding in the river valley Tourists visiting waterfall and the source of rivers Recreation activities
9.		 RIVER NIGER	✞ It flows through Niger and Nigeria, and pours its water into the Atlantic ocean. ✞ It forms a delta at its mouth. ie.the Niger delta. ✞ River Benue is the main tributary of river Niger. ✞ The construction of Kainji dam on river Niger resulted into formation of lake Kainji. <u>Importance of the Niger delta.</u>	Guided discovery Observation Brainstorming	To help the learner understand the main river in west Africa To help learners understand the importance of R. Niger in	Identifying the main tributaries of river Niger and River Volta	Generation of hydroelectricity Fetching water for domestic use Irrigation farming Using a bridge to cross

				✞ It has many oil wells. ✞ It has fertile soils for crop growing. ✞ It attracts tourists who bring in income.	Question and answer Guided group discussion Picture interpretation	irrigation and farming To help learners learn about Niger's role in settlement and early African kingdoms To help learner understand the river that forms lake Volta		Fishermen earning a living Using water to cool industrial machines Flooding in the river valley Tourists visiting waterfall and the source of rivers Recreation activities
			 RIVER VOLTA	✞ It flows through Burkina Faso and Ghana, and pours its water into the Atlantic ocean. ✞ The construction of the Akosombo power dam along river Volta resulted into creation of lake Volta. ✞ Lake Volta is the largest manmade lake in Africa. <u>Main tributaries of river Volta.</u> -Black Volta, - White Volta, - River Oti.				

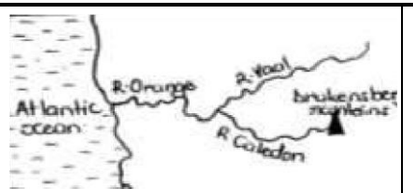
				one of the world's largest manmade lake To help learners learn about hydroelectric power generation at Akosombo dam To help leaners appreciate the how river Volta supports agriculture in Ghana		

10.	Identifies the countries River Niger flows through (Niger and Nigeria) States where River Niger pours its water (Atlantic Ocean). Mentions the main tributary of River Niger (River Benue). Lists the importance of the Niger Delta (oil, fertile soils, tourism). Explains what a delta is using the Niger Delta. Describes why the Niger Delta is important to the people living there.	RIVER ZAMBEZI  	✞ It flows through Zambia and Mozambique. ✞ It pours its water into the Indian ocean. ✞ It forms a delta at its mouth. ✞ Lake Kariba was formed after the construction of Kariba dam along river Zambezi. <u>Hydro electric power dams along river Zambezi.</u> ✞ Kariba dam in Zambia ✞ Cahora Bassa dam in Mozambique ✞ Kafue dam in Zambia NB:<u>Victoria falls</u> along river Zambezi in Zambia are the highest waterfalls in Africa.	Guided discovery Observation Brainstorming Question and answer Guided group discussion Picture interpretation	To enable the learner understands the river that hosts Victoria fall, a major world wonder To help learners recognize the role of river Zambezi in hydro power	Identifying the major tributaries of river Zambezi and river Tana	Generation of hydroelectricity Fetching water for domestic use Irrigation farming Using a bridge to cross Fishermen earning a living Using water to cool industrial machines Flooding in the river valley

	Locates River Niger and the Niger Delta on a map of Africa. Draws a simple diagram showing a river and a delta. States the countries drained by River Volta (Burkina Faso and Ghana). Mention the dam that led to the creation of Lake Volta (Akosombo Dam). Identifies the main tributaries (Black Volta, White Volta, River Oti). Explains why people build dams on rivers.	 RIVER TANA	✞ It originates from the Kenya highlands and flows into the Indian ocean. ✞ River Thiba and River Nyamindi are the main tributaries of river Tana. ✞ River Thiba supplies water to Mwea Tebere irrigation scheme. (famous for rice growing) ✞ The Seven Forks scheme was built on river Tana. <u>Power dams built along river Tana.</u> -Gitaru dam, -Kindaruma, -Kamburu, -Masinga dam, -Kiambere dam.		To help learners understand the longest river in Kenya To enable the leaner recognize the role of R. Tana in hydroelectric power generation To help learners understand how River Tana supports fishing and domestic water supply		Tourists visiting waterfall and the source of rivers Recreation activities

	Describes how dams help to create artificial lakes. Identifies the countries River Zambezi flows through (Zambia and Mozambique). State where River Zambezi pours its water (Indian Ocean). Mentions dams along River Zambezi (Kariba, Cahora Bassa, Kafue). States that Lake Kariba was formed after building Kariba Dam. Explains the importance of					

	waterfalls and dams for tourism and electricity. Describes what a delta is at the river's mouth. Locates River Zambezi, Kariba Dam, and Victoria Falls on a map. Identifies the origin of River Tana (Kenya Highlands). States where it pours its water (Indian Ocean). Mentions its main tributaries (River Thiba, River Nyamindi). States that River Thiba supports the					

	Mwea Tebere Rice Scheme. Lists power dams on River Tana (Gitaru, Kindaruma, Kamburu, Masinga, Kiambere). Explains why farmers use river water for irrigation.					
11.		 RIVER ORANGE	✞ It originates from the Drakensberg mountains and flows into the Atlantic ocean. ✞ River Vaal and river Caledon are the main tributaries of river orange. ✞ Vaal power dam and Verwoerd dam were built along river orange in South Africa.	Guided discovery Observation Brainstorming	To help learners understand the major river in southern Africa To help learners to learn about the importance of irrigation	Identifying the main tributaries of river Orange, River Limpopo	Generation of hydroelectricity Fetching water for domestic use Irrigation farming Using a bridge to cross

				Question and answer Guided group discussion Picture interpretation	especially in south Africa and Namibia To help learners recognize its role as a natural boundary between south Africa and Namibia To help learners understand one of key rivers flowing through Botswana, Mozambique, south Africa and Zimbabwe		Fishermen earning a living Using water to cool industrial machines Flooding in the river valley Tourists visiting waterfall and the source of rivers Recreation activities

RIVER LIMPOPO	✞ It flows into the Indian ocean. ✞ It forms a natural boundary between South Africa and Mozambique.

Rivers in Africa and the water bodies they pour their water.

Ocean/ sea/ lake	Rivers
Lake Victoria	-River Kagera -River Katonga -River Mara -River Nzoia -River Yala
Indian ocean	-River Ruvuma -River Rufiji -River Pangani -River Tana, -River Galana,
Atlantic ocean	-River Congo, -River Senegal, -River Niger, -River Gambia, -River Volta, -River Orange.
Mediterranean sea	-River Nile

12.	State the source of River Orange (Drakensberg Mountains). Mention where River Orange pours its water (Atlantic Ocean). Identify the main tributaries of River Orange (Vaal and Caledon rivers). Mention two dams built on River Orange (Vaal Dam, Verwoerd Dam). Explain why tributaries help to increase the volume of a river. Describe the importance of	**Importance of lakes and rivers** They help in rain formation. They act as fishing grounds. Rivers help in generation of hydro electricity. They attract tourists who bring in income. They form natural boundaries between countries. They are used for recreation. They are used as a medium of water transport. They provide water for irrigation, domestic and industrial use. **A diagram showing the formation of Convectional rainfall.** **Problems associated with lakes and rivers.** Drowning Flooding They harbour disease vectors. Rivers and lakes hinder road and railway construction **Problems facing lakes and rivers** Silting which reduces the depth of water bodies. Overuse of water from lakes and rivers. Drought which lowers the water level.	Guided discovery Observation Brainstorming Question and answer Guided group discussion	To enable learners appreciate lakes as fishing grounds To enable learners understand hydropower production To enable learners recognize trade and transport value To enable the learners appreciate lakes	Identifying the importance of lakes and rivers Drawing the diagram showing the formation of convectional rainfall	Water hyacinth blocking boats Cleaning fishing nets at the shores Brick making using water from lakes People learning to swim A community fetching water during dry season Celebrating cultural social festivals

	building dams along rivers. Locate River Orange and its tributaries on a map of Africa. State where River Limpopo pours its water (Indian Ocean). Identify that River Limpopo forms a natural boundary between South Africa and Mozambique. Explain the meaning of a natural boundary using River Limpopo. Describe why rivers can act as borders between countries.	Dumping of wastes into water bodies. Over fishing **Why most parts of rivers in Africa are not navigable (not used for transport)** Some rivers have waterfalls and rapids. Some rivers have dangerous aquatic animals. Some parts of rivers are shallow and narrow Some rivers have floating vegetation. (suds)		and rivers as natural boundaries between countries To help the learner learn about wildlife survival		Tourists taking boat rides Floating fish cages A child observing his/her reflection Traders loading goods on to boats Using clay for pottery Children playing with reeds

	Locate River Limpopo on a map of Southern Africa. Draw a simple sketch showing a river forming a boundary. Identify major African rivers and the lakes, seas, or oceans they flow into. State examples of rivers that pour into Explain why rivers flow into larger water bodies. Compares rivers based on the oceans or lakes they pour into.					
	Identifies physical features such as mountains, plains, plateaus, lakes,	**THE INFLUENCE OF PHYSICAL FEATURES** <u>On People</u>	Guided discovery	To help learners understand how physical	Describing the influence of	Mountain areas and temperature

	rivers, and rift valleys. States how physical features influence human settlement. Mentions why few people live in high mountains (cold temperatures, poor transport). Explains why people prefer settling in areas with fertile soils. Describes how steep slopes affect transport in highland areas. Explains why building houses and roads is easier on plateau areas.	Mountainous areas attract human settlement due to the presence of fertile soils for cultivation. Plains favour pastoralism due to the presence of pasture. Ground transport is poor in highland areas due to the presence of steep slopes. Many people settle around lakes and rivers because such areas receive reliable rainfall and have fertile soils which support crop cultivation. **Factors which attract human settlement in;** **i) Plateau areas** Presence of pasture for animals. Agricultural mechanisation is easy in such areas. Construction of houses, roads, railways and industries is easy. Natural hazards like floods are not common in these areas. **ii) Highland areas** Presence of fertile soils. Availability of reliable rainfall. **Common economic activities carried out in highland areas.**	Observation Brainstorming Question and answer Guided group discussion Role play Debate	determine settlement patterns To help learners know why people choose certain occupations based on features around them To enable learners explain how physical features affect climate, transport routes and	physical features	Settlement along river banks Highlands and farming Fertile volcanic soils Swamps as grazing areas Flooding near rivers Using hills for communication masts Basin areas supporting large farms Mountains limiting transport

	Compares settlement patterns in plains, mountains, and lake areas. Draws and labels physical features (mountains, plateaus, plains).State that plants grow well where there is enough rainfall and fertile soil. Identify places with many plants (lake shores, mountain slopes). Identify places with few plants (mountain tops). Explains why rainfall and fertile soils support plant growth.	Crop farming Mining Dairy farming **Reasons why there are very few people living in higher parts of mountains.** Transport is poor in higher parts of the mountains Highland areas experience very low temperatures which do not favour human settlement. **b) On Plants** Areas around lakes and rivers have many plants due to plenty of rainfall received. Slopes of mountains have forests due to presence of fertile soils and reliable rainfall received. Mountain tops have few plants due to low temperatures. c) <u>On Animals.</u> Plains have plenty of grass which attracts animals. Water bodies favour aquatic animals. Pastoralism is common on plateau areas due to the presence of pasture for animals.		economic activities To help learners recognize how physical features shape culture, traditional practices and way of life To help learners appreciate the need for conserving physical features to support future generations		Traditional houses built with local materials Pastoralists following grassland Rivers used in electricity Drought in lowland areas Mountain slopes used for farming Waterfalls attracting

	Describes how temperature at different altitudes affects plant growth. Identifies places where animals live (plains, lakes, highlands, forests). States that plains support grazing animals due to plenty of grass. Mentions that water bodies support aquatic animals. Identifies mountain animals like mountain gorillas. Explains why pastoralism is common in plateau areas	There are many climbing animals eg. mountain gorillas in highland areas due to the presence of many fruit trees. **d) On Climate.** Mountains receive plenty of relief rainfall. The wind ward side of a mountain receives plenty of rainfall because it receives warm moist air. The lee ward side of the mountain receives very little rainfall because it receives dry winds. Mountainous areas experience very low temperatures because they lie at a higher altitude. Rift valleys and low lands experience very high temperature because they are of low altitude. Lake shores receive plenty of convectional rainfall.				

	(availability of pasture). Describes how vegetation influences the type of animals found in an area. State that mountains receive relief rainfall. Identify the windward side as wetter and leeward side as drier. Mentions that highlands are cooler than lowlands. State that lakeshores receive convectional rainfall. Explains how altitude affects					

	temperature (high altitude = cold). Describe how mountains influence rainfall patterns. Explains how lakes cause heavy rainfall.					